

Customer Behavior Analytics Pipeline

A robust architectural flow for categorizing financial transactions, aggregating attributes, and computing dynamic behavior metrics.

Phase 1: Data Ingestion

Financial Transactions

The primary high-volume data stream containing raw records of customer spending, transfers, and point-of-sale activities.

Key attributes parsed:

- Raw text descriptions
- Transaction amounts & timestamps
- Merchant identification codes

Deposit Records

Periodic snapshots detailing account funding, direct deposits, income streams, and liquidity events.

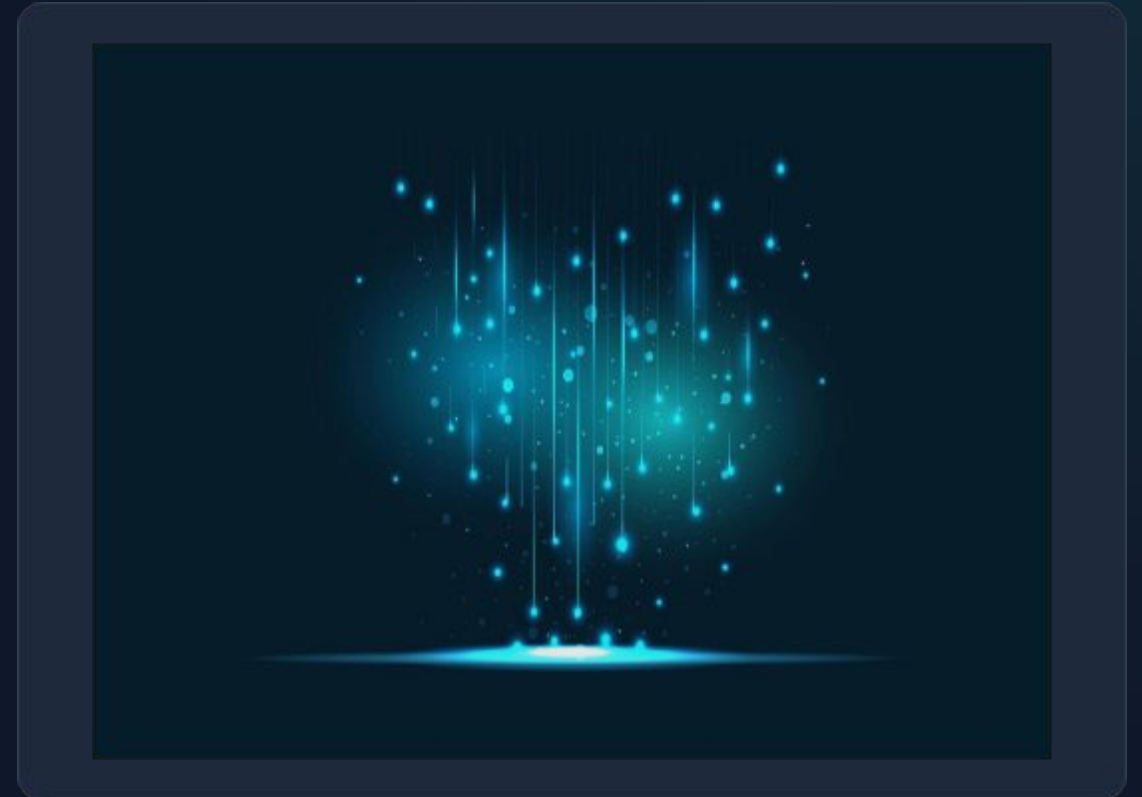
Key attributes parsed:

- Inbound fund origins
- Funding frequency (recurring vs. ad-hoc)
- Deposit magnitude & variation

Phase 2: Transaction Categorization

Raw transaction descriptions are processed through a rules-based engine places each transaction into a distinct category

- ✓ **Pattern Recognition:** Identifies whether or not a transaction is recurring based on pattern recognition.
- ✓ **Standardization:** Maps transactions to definitive categories (e.g., Groceries, Utilities, Entertainment, Rent).
- ✓ **Anomaly Detection:** Flags unusual spending patterns that deviate from established customer history (as potential fraud).
- ✓ **Data Enrichment:** Prepares cleanly tagged data for the subsequent aggregation layer.



Phase 3: Parallel Aggregation & Joining



Transaction Rollups

Categorized spending is grouped by month to compute average velocity and volume.



Aggregation

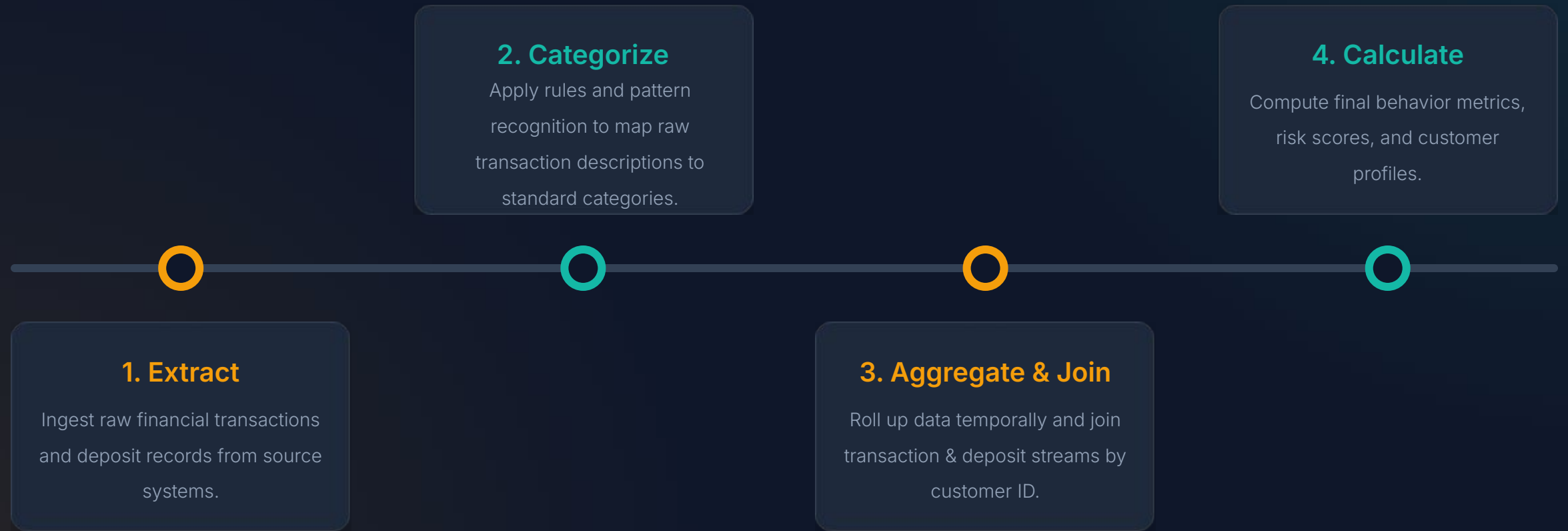
Transaction attributes (e.g., monthly income, debt payments) are aggregated along with deposit account balances to calculate behavioral metrics (e.g., debt-to-income ratios).



Feature Joining

The aggregated transaction metrics and deposit summaries are unified via customer identifier to create a single master record.

End-to-End Pipeline Flowchart



Project Implementation & Impact

I developed this workflow for a recent project, achieving unprecedented data processing velocity to power advanced financial modeling.



Powered by PySpark

Engineered using Apache Spark's Python API to enable distributed, high-performance data transformations across cluster nodes.



Massive Scale

Optimized algorithms allowed the pipeline to ingest, categorize, and aggregate billions of transactions in just a few hours.



Downstream Modeling

The pipeline outputs directly supply critical behavioral features for risk rating algorithms and credit product opportunity modeling.

Phase 4: Behavior Metrics Output

360°
CUSTOMER PROFILE

Synthesizing Actionable Intelligence

The final output layer computes sophisticated behavioral metrics by correlating the aggregated spending habits against accumulated deposit liquidity.

These calculated metrics can power downstream applications, enabling personalized financial product recommendations, dynamic risk scoring, and targeted customer engagement strategies.

Image Sources



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